

Thang Pang Ern

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Education

National University of Singapore

Aug 2024 -

BSc. Hons in Pure Mathematics (Highest Distinction)

- ▷ **Coursework:** Discrete Mathematics, Single and Multivariable Calculus, Linear Algebra, Real Analysis, Numerical Analysis, Probability Theory, Complex Analysis, Ordinary Differential Equations, Mathematical Modelling, Analytic Number Theory, Differential Geometry
- ▷ Dean's List for AY25/26 Semester 2

École Polytechnique Fédérale de Lausanne (EPFL)

Sept 2026-Jan 2027

Bachelor's Degree in Mathematics (exchange student)

- ▷ **Coursework:** Non-Linear Programming, Partial Differential Equations, Topology, Linear Optimisation, Analytic Number Theory

Nanyang Junior College

Feb 2020 - Dec 2021

Singapore-Cambridge GCE A-Level

- ▷ **Coursework:** H2 Mathematics, H2 Further Mathematics, H3 Mathematics
- ▷ 6 Distinctions

Experience

Optiver

Aug 2026

Quantitative Trading and Research

- ▷ FutureFocus Program
- ▷ Invited to Exclusive Insights Day program

PreciX

Oct 2025-Mar 2026

Algorithm Engineer

- ▷ Worked on knee biomechanics problems and implemented matrix-based geometric computation pipelines in Python, emphasising on numerical stability
- ▷ Developed and integrated spherical linear interpolation (SLERP) for smooth rotation/orientation transitions (quaternion/rotation matrix workflows)

NUS Department of Mathematics

Jul 2025-May 2026

Teaching Assistant

- ▷ Taught MA2001 Linear Algebra I as part of the Undergraduate Teaching Opportunities Programme (UTOP)
- ▷ Prepared comprehensive written solutions and video explanations to tutorial questions

NUS Mathematics Society

May 2025-Jul 2026

President and Project Director

- ▷ Organised Problematic 2025, a faculty-wide mathematics competition and co-organised Sudoku Contest 2025, a faculty-wide puzzle competition
- ▷ Launched a professor sharing series on research and academic experiences
- ▷ Recruited speakers for a senior-junior sharing session on research, careers, and graduate studies
- ▷ Invited to London for the final round of the UK University Integration Bee

- ▷ Taught Mathematics: Analysis and Approaches (AA), Cambridge International AS & A Level Mathematics and Further Mathematics, Pearson Edexcel A Level Further Mathematics and STEP

Publications and Preprints

1. **Linear Algebra: A Balance of Visualisation and Abstraction**, with D. W. Wangsa, Joel Q. L. Chang, T. L. Kho
To appear in Springer
2. **Cutting the Gamma Function in Half**
To appear in *Parabola*, Volume 63, Issue 1
3. **The Three Square Geometry Problem and an Extension**
To appear in *Parabola*, Volume 62, Issue 3
4. **Inequivalent Arithmetic Expressions on a Multiset of Operands**, in preparation
5. **On the Limiting Density of a gcd Map**, submitted
6. **The Irrationality of $\zeta(3)$ using Modular Forms**, preprint
arXiv:2607.17123 [↗](#)
7. **On the Ross-Littlewood Paradox**, submitted
8. **Proof without Words: Geometric Evaluation of a Limit**, submitted
9. **Some Generalizations of the Bridge and Torch Problem**, with G. Sayson
arXiv:2608.05190 [↗](#)
10. **The Area of a Triangle**, submitted
11. **A Proof of Ramanujan's Classic π Formula**, with D. W. Wangsa
arXiv:2411.15803 [↗](#)

Talks

The Area of a Triangle (Slides) ↗ <i>Future Generations of Scientists (FGS) Camp 2026</i>	Jul 2026
The Integers: Simple yet Complicated <i>Dunman High School</i>	Apr 2026
Pólya's Random Walk Theorem (Slides) ↗ <i>National University of Singapore, Mathematics Seminar</i>	Feb 2026
Weyl Sums and its Applications (Slides) ↗ <i>National University of Singapore, UROPS Sharing Session</i>	Oct 2025
The Life and Legacy of Srinivasa Ramanujan (Slides) ↗ <i>National University of Singapore, Mathematics Seminar</i>	Oct 2025
The Art and Analysis of Assumptions (Slides) ↗ <i>Future Generations of Scientists (FGS) Camp 2025</i>	Jul 2025
The Prime Number Theorem (Slides) ↗ <i>National University of Singapore, Mathematics Seminar</i>	Mar 2025
Bridge Crossing (Slides) ↗ <i>Future Generations of Scientists (FGS) Camp 2024</i>	Aug 2024
Simple yet Complicated Numbers (Slides) ↗	Aug 2023

Technologies

Languages: Python, Java, HTML, CSS, MATLAB, L^AT_EX, Mathematica

Technologies: Microsoft Office, VEGAS Pro, Premiere Pro, After Effects, Illustrator, Photoshop, Lightroom

Contributions to the Online Encyclopedia of Integer Sequences (OEIS)

A396925:  Numbers k which have a divisor d such that $d^2 < k$ and $\phi(d) = \phi(k/d)$, where ϕ denotes Euler's totient function. Aug 2026

A195055:  Decimal expansion of $\frac{\pi^2}{3}$. Jul 2026
 Contributed the following property: equals

$$\sum_{n=1}^{\infty} \frac{H(n)}{T(n)},$$

where $H(n)$ is the n^{th} harmonic number and $T(n)$ is the n^{th} triangular number.

A062119:  $a(n) = n!(n-1)$. Jul 2026
 Contributed the following property: equals

$$\left(\sum_{k=1}^{\infty} \prod_{i=1}^n \frac{1}{k+i} \right)^{-1}.$$

A397310:  Decimal expansion of $\int_0^{\pi} \cos(\cos x) dx$. Jul 2026

A069998:  Decimal expansion of $\sqrt{\frac{\pi}{2}}$. Jul 2026

Contributed the following property: equals $\int_0^{\infty} \frac{\sin(x^2)}{x^2} dx$.

A039686:  Squares which are the concatenation of two nonzero squares. May 2026
 Contributed the following property: the sequence is infinite because it contains $49 \cdot 10^{2i}$ for $i \geq 0$ since $(7 \cdot 10^i)^2$ can be split into the initial 4 and $9 \cdot 10^{2i}$ with square root $3 \cdot 10^i$.

A395338:  Primes whose square is a concatenation of exactly 2 distinct strictly increasing non-zero squares. Apr 2026

A342316:  Decimal expansion of $\frac{\pi}{2} - \log 2$. Feb 2026
 Contributed the integral representation

$$\int_1^{\infty} \log \left(1 + \frac{1}{x^2} \right) dx.$$

A392834:  Time taken to get n people from one side of a bridge to the other where (a) the only flashlight must be carried when crossing; (b) at most three people may cross at the same time; (c) a group crosses at the speed of the slowest member; and (d) the k^{th} person requires k seconds to cross the bridge. Jan 2026

With Gerard Sayson

A392835:  Number of ordered pairs (i, j) with $1 \leq i, j \leq n$ such that $\gcd(i + j, ij) = \gcd(i, j)$

Jan 2026

With Loh Wei Xuan Ryan and Malcolm Tan Jun Xi